THE THESIS TITLE

BY

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## ABSTRACT

This thesis presents a template for graduate theses and dissertations at the University of Rhode Island. The template uses the `rhode-island-thesis` document class, which enforces the formatting requirements of the URI Graduate School while automating the generation of all preliminary pages.

The thesis class supports both master's theses and PhD dissertations, per-chapter or single reference lists, numeric and author–year citation styles, and a variety of committee sizes. Detailed instructions for compiling the document and running BibTeX are provided in the comments of the root `rhode-island-thesis.tex` file.

Results demonstrate that the template compiles cleanly under a standard TeX Live installation using `pdflatex` and `bibtex`, and that it produces correctly formatted preliminary pages, chapter headings, figures, tables, and reference lists conforming to URI Graduate School guidelines.

## ACKNOWLEDGMENTS

I would like to express my sincere gratitude to my advisor, Dr. Jane Smith, for their guidance, patience, and invaluable support throughout this research.

I am also deeply grateful to the members of my thesis committee, Dr. Robert Johnson and Dr. Maria Garcia, for their insightful feedback and encouragement.

This work was supported in part by [funding source or grant number]. I thank my colleagues in the [Lab/Group Name] for many stimulating discussions and for making the lab a welcoming place to work.

Finally, I am grateful to my family for their unwavering support and encouragement throughout my graduate studies.

## DEDICATION

*To my family,*

*for their love and support.*

# PREFACE

The preface is optional. Use it to describe the origin, purpose, or scope of the work, or (in manuscript format) to explain how the manuscripts relate to one another. Delete the `\preface` line in `rhode-island-thesis.tex` if you do not need one.

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# MANUSCRIPT 1

# Introduction

## 1.1 Background and Motivation

Provide background information that motivates your research. Explain the context of the problem and why it is important. This section should be accessible to a reader who is knowledgeable in the general field but may not be familiar with the specific topic.

A typical thesis introduction will survey the relevant literature [1] and identify the gap in knowledge that this work addresses [2]. Each claim that draws on prior work should be supported by a citation.

## 1.2 Problem Statement

State the specific research problem clearly and concisely. The problem statement should flow naturally from the motivation established in Section 1.1.

- What is not yet known or understood?
- Why does that gap matter?
- What is the scope of this work?

## 1.3 Research Objectives

The objectives of this thesis are:

1. To demonstrate the structure of a URI thesis template.
2. To show how figures, tables, and equations are included.
3. To illustrate the citation and bibliography workflow.

## 1.4 Contributions

The main contributions of this work are summarized below.

1. A compilable, fully annotated thesis template conforming to URI Graduate School requirements.

2. Worked examples of figures, tables, equations, and cross-references.

3. Documentation of the BibTeX workflow for both per-chapter and single-list reference modes.

## 1.5 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 provides a detailed review of related work. Chapter 3 describes the methodology and experimental setup. Conclusions and directions for future work are presented in Chapter 3. Supplementary material appears in the appendix.

## List of References

[1] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Reading, MA: Addison-Wesley, 1994.

[2] D. E. Knuth, *The TeXbook*. Reading, MA: Addison-Wesley, 1986.

# MANUSCRIPT 2

## Background and Related Work

### 2.1 Related Work

This chapter reviews prior work relevant to the thesis topic. A thorough literature review situates your contribution within the existing body of knowledge and justifies your methodological choices.

Each paragraph should synthesize related sources rather than summarizing individual papers in isolation [1]. Group sources thematically:

- Approach A and its variants [2].
- Approach B and its limitations [3].
- Hybrid methods that combine elements of both [1].

### 2.2 Including Figures

Figures are included with the standard `figure` environment. The class automatically adds them to the List of Figures. Always include a descriptive caption and a `\label` so the figure can be cross-referenced.

Figure 1 shows a placeholder rectangle that should be replaced with your actual figure file (`.pdf`, `.png`, or `.eps`).

When a figure is wider than the text column, the `[width=\textwidth]` option scales it to fit. Never let a figure extend into the margin.

### 2.3 Including Tables

Tables should use the `booktabs` package for clean horizontal rules, as shown in Table 1. Vertical rules are generally discouraged in professional typesetting.

The `[htbp]` placement specifier asks LaTeX to place the float *h*ere, at the *t*op of a page, at the *b*ottom, or on a separate *p*age of floats, in that order of preference.



Figure 1. Longer caption that appears below the figure. The short caption in brackets is what appears in the List of Figures. If only one argument is given it is used in both places.

Table 1. A comparison of example methods. Replace this with your data.

| Method | Reference | Accuracy (%) | Runtime (s) |
|---|---|---|---|
| Approach A | [2] | 91.2 | 4.3 |
| Approach B | [3] | 88.7 | 2.1 |
| Approach C | [1] | 93.5 | 8.9 |
| This work | — | **95.1** | 5.7 |

### 2.4 Equations

Inline mathematics is set with dollar signs: $E = mc^2$. Displayed equations are set with the `equation` environment:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{1}$$

Equation (1) shows a Gaussian probability density function with mean $\mu$ and standard deviation $\sigma$. For aligned multi-line derivations, use the `align` environment:

$$a + b = c, \tag{2}$$

$$d \cdot e = f. \tag{3}$$

Equations (2) and (3) illustrate aligned equations. Equation numbers are set sequentially by the class (or per-chapter if the `nonsequential` option is used).

### List of References


[1] H. Kopka and P. W. Daly, *A Guide to LaTeX 2ε*, 3rd ed. Harlow, England: Addison-Wesley, 1999.

[2] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Reading, MA: Addison-Wesley, 1994.

[3] D. E. Knuth, *The TeXbook*. Reading, MA: Addison-Wesley, 1986.

# MANUSCRIPT 3

## Methodology, Results, and Conclusions

### 3.1 Methodology

Describe your experimental or analytical methods in enough detail that a reader with the necessary background could reproduce your work. Reference standard techniques with citations [1] and reserve detailed derivations for appendices if they would interrupt the narrative flow.

#### 3.1.1 Experimental Setup

Describe your equipment, software, datasets, and any parameters that affect reproducibility. If your setup involves multiple stages, a numbered list or a flow-diagram figure helps the reader follow the sequence.

#### 3.1.2 Evaluation Metrics

Define the metrics you use to evaluate your results. Quantitative metrics should be accompanied by their mathematical definitions, as in Equation (4):

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}. \tag{4}$$

### 3.2 Results and Discussion

Present your results objectively, then interpret them. Connect each result back to the objectives stated in Section 1.3.

When comparing to prior art (Table 1 in Chapter 2), be precise about the conditions under which each method was evaluated.

### 3.2.1 Quantitative Results

Summarize numerical results in tables, and use figures to reveal trends or distributions that a table cannot capture clearly.

### 3.2.2 Discussion

Discuss the implications of your results. What do they tell you about the research question posed in Chapter 1? Where do the results agree with prior work, and where do they differ?

## 3.3 Conclusions

State the conclusions that can be drawn from your work. Each conclusion should follow directly from a result or observation presented in Section 3.2.

1. The template compiles cleanly under a current TeX Live using `pdflatex` and `bibtex`.

2. Per-chapter and single-list reference modes both function correctly with the provided `.bst` files.

3. The preliminary pages (title, approval, abstract, TOC, List of Figures, List of Tables) are generated automatically.

## 3.4 Future Work

Identify open questions and directions for future investigation. Future work might include:

- Extension of the methodology to additional domains.

- Investigation of alternative approaches not explored here.

- Long-term evaluation studies.

## List of References

[1] H. Kopka and P. W. Daly, *A Guide to LaTeX 2ε*, 3rd ed. Harlow, England: Addison-Wesley, 1999.

# APPENDIX

## Supplementary Material

This appendix contains material that supports the main text but would interrupt the flow of the argument if placed in the body of the thesis. Typical appendix content includes:

- Detailed mathematical derivations.
- Extended data tables.
- Pseudocode or source code listings.
- Instrument specifications or calibration data.
- Copies of survey instruments or protocols.

### A.1 Detailed Derivation

Place extended derivations here. For example, the derivation of Equation (4) from first principles, which would have been too long to include in Chapter 3.

### A.2 Additional Data Tables

Additional tables that did not fit in the main chapters can appear here. Figures and tables in appendices are excluded from the List of Figures and List of Tables by the class, as required by the Graduate School.